

Assign 2: Construct Binary Tree

Construct a binary tree by using the last 5 digits of your student's id as the post-order sequence include repeating numbers and zeros. Create the in-order sequence by move the last digit to the second digit and swap the fourth and fifth digits at the end.

Write the output in pre-order traversal.

For Example: if student's id is 6706021 632032

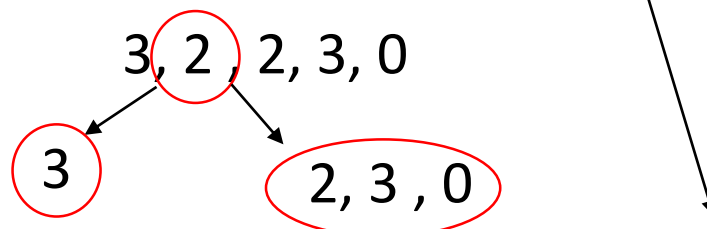
then the last 6 digits are 32032

So the post-order sequence is 3, 2, 0, 3, 2

The in-order sequence is 3, 2, 2, 3, 0

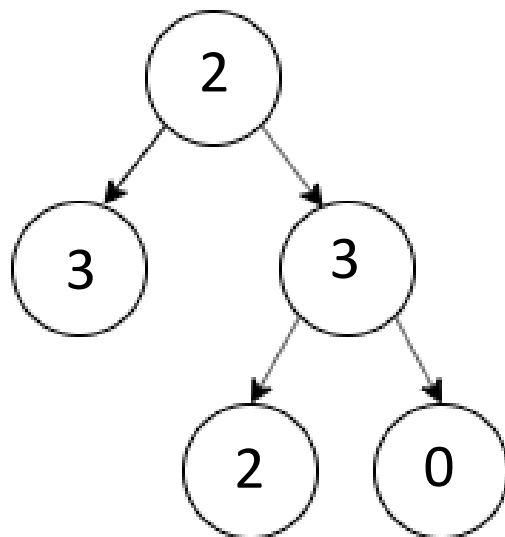
Hint 1: Root node of tree is the last visiting node in Post-order traversal. Thus, Root Node = 2.

Hint 2: With the in-order sequence, we can construct binary tree from hint1



The second last node in post-order traversal is 3. Thus, node 2 becomes left child of node 3 and node 0 becomes right child of node 3. Thus, the final tree is

Result:



Output is : ?